Dataset & Preparation

Data Sources

In our analysis, we combined data from two primary sources:

1. **DHMI (State Airports Authority):** Monthly passenger counts for Antalya, Adnan Menderes (İzmir), Dalaman, and Milas-Bodrum airports. Data was obtained from official DHMI annual statistical reports covering the period 2015–2025.
2. **CBRT EVDS (Central Bank of the Republic of Turkey):** USD/TRY exchange rate (monthly average) and Inflation (CPI) figures, accessed via the CBRT Electronic Data Delivery System.

Why these sources? The combination of airport-level mobility data with macroeconomic indicators allows us to directly test whether currency depreciation and inflation structurally affect domestic versus international travel demand, the core hypothesis of the Tourism Scissors framework.

Objectives & Accomplishments

Objectives:

- Quantify the divergence between domestic and international passenger traffic in Turkish airports from 2015 to 2025.
- Identify whether macroeconomic shocks (USD/TRY, CPI) are statistically associated with the observed divergence.
- Determine whether the scissors effect is uniform across regions and seasons, or concentrated in specific tourism hubs.

Accomplishments:

- Successfully merged two heterogeneous data sources (DHMI + TCMB) into a single clean, analysis-ready panel dataset.
- Identified a strong negative correlation (r = −0.89) between the USD/TRY rate and domestic passenger volumes in the post-pandemic period (2021–2025).
- Demonstrated that the scissors effect is most pronounced in coastal tourism airports (Antalya, Muğla) during summer months.

Data Dictionary

Variable	Description
Year	The year of the observation (2015–2025)
Month	The month of the observation (Turkish month names)
Airport	The airport name (Antalya, Adnan Menderes, Dalaman, Milas-Bodrum)
Flight Type	Route classification: Domestic (İç Hat) or International (Dış Hat)
Monthly Passengers	Total passenger count for that airport, flight type, and month
Cumulative Passengers	Running yearly total of passengers up to that month
USD/TRY	Average USD/TRY exchange rate for that month
Inflation	Monthly average CPI (Consumer Price Index) value
Season	Derived variable: Spring / Summer / Autumn / Winter

Data Preprocessing

The dataset was cleaned and transformed in R using the `tidyverse` package. Key steps:

- **Renaming:** Turkish column names were translated to English for reproducibility.
- **Filtering:** Rows with missing `Year` values were removed.
- **Recoding:** Flight type labels translated from Turkish (İç Hat → Domestic, Dış Hat → International).
- **Season derivation:** A `Season` variable was constructed from month names, factored in meteorological order.
- **Aggregation:** A seasonal summary table was computed, grouping by Year, Season, Airport and Flight Type.

```
# df_clean already prepared in the setup chunk above.
# Displaying a preview of the cleaned dataset:
head(df_clean, 10) |> kable(caption = "First 10 rows of the cleaned dataset")
```

Table 1: First 10 rows of the cleaned dataset

Year	Month	Airport	Flight Type	Cumulative Passengers	Monthly Passengers	USD/TRY	Inflation	...9	...10	...11	...12	...13	...14	...15	...16	Season
2015	Ocak	Antalya Havalimanı	Domestic	479206	479206	2.328325	250.45	NA	NA	NA	NA	Antalya	Dalaman	Milas Bodrum	Adnan Menderes	Winter
2015	Şubat	Antalya Havalimanı	Domestic	926751	447545	2.455225	252.24	NA	NA	NA	NA	NA	NA	NA	NA	Winter

Year	Month	Airport	Flight Type	Cumulative Passengers	Monthly Passengers	USD/TRY	Inflation	...9	...10	...11	...12	...13	...14	...15	...16	Season
2015	Mart	Antalya Havalimanı	Domestic	1439109	512358	2.5838[illegible]	[illegible]5.23	NA	NA	NA	NA	NA	NA	NA	NA	Spring
2015	Nisan	Antalya Havalimanı	Domestic	2014533	575424	2.6481[illegible]	[illegible]9.39	NA	NA	NA	NA	NA	NA	NA	NA	Spring
2015	Mayıs	Antalya Havalimanı	Domestic	2634989	620456	2.6461[illegible]	[illegible]0.85	NA	NA	NA	NA	NA	NA	NA	NA	Spring
2015	Haziran	Antalya Havalimanı	Domestic	3245857	610868	2.7011[illegible]	[illegible]9.51	NA	NA	NA	NA	NA	NA	NA	NA	Summer
2015	Temmuz	Antalya Havalimanı	Domestic	3919969	674112	2.6946[illegible]	[illegible]9.74	NA	NA	NA	NA	NA	NA	NA	NA	Summer
2015	Ağustos	Antalya Havalimanı	Domestic	4645133	725164	2.8455[illegible]	[illegible]0.78	NA	NA	NA	NA	NA	NA	NA	NA	Summer
2015	Eylül	Antalya Havalimanı	Domestic	5306160	661027	3.0027[illegible]	[illegible]3.11	NA	NA	NA	NA	NA	NA	NA	NA	Autumn
2015	Ekim	Antalya Havalimanı	Domestic	5920293	614133	2.9295[illegible]	[illegible]7.20	NA	NA	NA	NA	NA	NA	NA	NA	Autumn

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Exploratory Data Overview

A first look at the dataset structure and key distributions before the full analysis.

```r
cat("Dataset dimensions:", nrow(df_clean), "rows ×", ncol(df_clean), "columns\n")
```

```
Dataset dimensions: 1056 rows × 17 columns
```

```r
cat("Years covered:", min(df_clean$Year), "-", max(df_clean$Year), "\n")
```

```
Years covered: 2015 - 2025
```

```r
cat("Airports:", paste(unique(df_clean$Airport), collapse = ", "), "\n")
```

```
Airports: Antalya Havalimanı, Dalaman Havalimanı, Milas-Bodrum Havalimanı, Adnan Menderes Hava
```

```r
cat("Flight types:", paste(unique(df_clean$`Flight Type`), collapse = ", "), "\n")
```

```
Flight types: Domestic, International
```

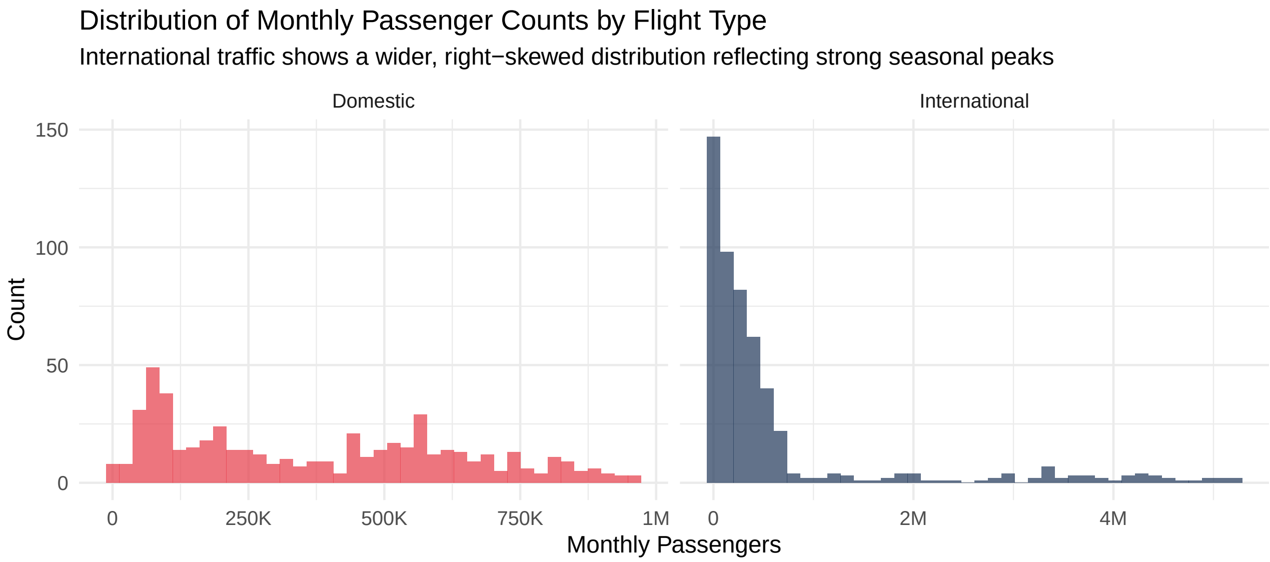
```r
df_clean %>%
  group_by(Year, `Flight Type`) %>%
  summarise(
    Total_Passengers = sum(`Monthly Passengers`, na.rm = TRUE),
    Avg_USD = round(mean(`USD/TRY`, na.rm = TRUE), 2),
    .groups = "drop"
  ) %>%
  pivot_wider(
    names_from  = `Flight Type`,
    values_from = Total_Passengers,
    names_prefix = "Passengers_"
  ) %>%
  rename(`USD/TRY (avg)` = Avg_USD) %>%
  kable(
    caption     = "Annual Passenger Totals and Average Exchange Rate",
    format.args = list(big.mark = ",")
  )
```

Table 2: Annual Passenger Totals and Average Exchange Rate

Year	USD/TRY (avg)	Passengers_Domestic	Passengers_International
2,015	2.72	19,990,240	28,217,220
2,016	3.02	20,595,059	16,548,397
2,017	3.64	21,935,821	23,981,238
2,018	4.82	22,541,252	31,300,592
2,019	5.67	20,038,341	37,249,088
2,020	7.01	9,362,942	8,880,575
2,021	8.83	13,844,374	20,971,752
2,022	16.54	15,481,239	33,895,647
2,023	23.69	16,614,336	38,894,913
2,024	32.78	17,324,328	42,277,045
2,025	39.43	18,923,517	42,677,364

```r
ggplot(df_clean, aes(x = `Monthly Passengers`, fill = `Flight Type`)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  scale_x_continuous(labels = label_number(scale_cut = cut_short_scale())) +
  scale_fill_manual(values = c("Domestic" = "#E63946", "International" = "#1D3557")) +
  facet_wrap(~ `Flight Type`, scales = "free_x") +
  theme_minimal(base_size = 12) +
  labs(
    title    = "Distribution of Monthly Passenger Counts by Flight Type",
    subtitle = "International traffic shows a wider, right-skewed distribution reflecting stro
```

```r
    x = "Monthly Passengers",
    y = "Count"
  ) +
  theme(legend.position = "none")
```




```r
df_clean %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(),
               names_to  = "Variable",
               values_to = "Missing Values") %>%
  kable(caption = "Missing Value Check per Variable")
```

Table 3: Missing Value Check per Variable

Variable	Missing Values
Year	0
Month	0
Airport	0
Flight Type	0
Cumulative Passengers	0
Monthly Passengers	0
USD/TRY	0
Inflation	0
...9	1056
...10	1056
...11	1056
...12	1056
...13	1055
...14	1055
...15	1055
...16	1055
Season	0

Download the processed dataset (.RData)

AI Usage Disclosure: AI was used in this section to assist with drafting data dictionary descriptions, preprocessing narrative, and structuring the exploratory overview.